

The Hidden World of Trade Credit: The Flexibility Role of Late Payments*

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Trade credit is a dominant source of short-term financing in supply chains, yet little is known about how firms repay trade credit in practice because data on realized payment timing are rarely available. Using a unique dataset from Dun & Bradstreet that contains supplier-reported late payment records, we examine how firms manage payment timing in their supply chain relationships. We propose and empirically support a framework that distinguishes between flexibility-driven and distress-driven payment delays. Firms with greater downstream cost-shifting needs, proxied by longer accounts receivable days, are more likely to delay payments to suppliers; this association is primarily observed among financially healthy firms. Payment delays are typically short-lived, whereas persistent delays are uncommon and more likely to be associated with supplier enforcement actions such as litigation or relationship termination. Operational outcomes also differ: late payments are associated with lower profitability, while financially healthy firms maintain stable inventory turnover but distressed firms exhibit higher turnover. These findings highlight the flexibility role of payment timing in trade credit and its implications for supply chain financing.

Keywords: trade credit, late payment, working capital management, supply chain finance

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1. Introduction

Trade credit has garnered significant attention in recent literature on the operations-finance interface (Birge 2014, Babich and Kouvelis 2018, Osadchiy et al. 2025, Zhang et al. 2026). By allowing buyers to receive goods or services before making payments, trade credit functions as a major source of short-term financing within supply chains (Haley and Higgins 1973, Yang and Birge 2018). A substantial body of work demonstrates how trade credit improves supply chain coordination by influencing production, ordering, and inventory decisions (Babich and Tang 2012, Kouvelis and Zhao 2012, Chod 2017). Empirical studies further highlight the economic importance of trade

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credit relative to bank financing. For instance, [Rajan and Zingales \(1995\)](#) found that accounts payable constituted 15% of total assets for US firms, while short-term loans accounted for only 7.4%, indicating that trade credit is a dominant source of short-term financing for many firms.

However, despite this extensive literature, our understanding of how firms actually repay trade credit in practice remains limited. Existing theoretical models typically assume either strict adherence to contractual payment terms or complete default, while empirical studies have predominantly focused on the supply side of trade credit—that is, how suppliers provide financing to their customers. Little is known about the payment behavior of downstream customers, largely because detailed data on realized payment timing have historically been unavailable.

Prior work often interprets late payment primarily through the lens of financial distress and systemic credit risk ([Boissay and Gropp 2013](#)). Yet payment timing may also serve as a flexible operational tool through which firms manage liquidity or as a channel through which firms exercise market power over suppliers. One indication that contractual payment terms may not fully capture actual payment behavior arises from the interest rates implied by standard trade credit contracts. For example, terms such as “2/10 Net 30” offer a 2% discount if payment is made within 10 days, with the full amount due within 30 days. If buyers forgo the discount and instead pay on the thirtieth day, the implied annualized interest rate exceeds 40%, which appears implausibly high relative to the typical cost of capital faced by most firms. One explanation is that firms often delay payment beyond the contractual due date, lowering the effective financing cost implied by the contract. Using our data, we estimate that the average payment delay reduces the effective interest rate implied by “2/10 Net 30” contracts from roughly 43.5% to approximately 15.6%.¹ This discrepancy highlights the importance of examining realized payment behavior rather than assuming strict adherence to contractual terms.

These observations raise several questions about how firms manage trade credit payments in practice. What drives payment delays beyond contractual terms? Do such delays reflect discretionary liquidity management, financial distress, or market power? How persistent are these delays, and how do suppliers respond to them? Finally, what are the operational consequences of delayed payments for inventory management and profitability? Addressing these questions is essential for understanding the operational role of trade credit within supply chains and for helping managers assess whether a customer’s delayed payment behavior reflects prudent liquidity management or heightened payment risk.

¹ The D&B data classify late payments into duration bins rather than exact days. We approximate each bin by its midpoint (15, 45, 75, 105, and 150 days for the 1–30, 31–60, 61–90, 91–120, and 121–180 day bins) and conservatively assign 180 days to the open-ended >180-day bin, then take the experience-weighted average across bins within each firm-year. The resulting sample average delay is approximately 30 days. The implied annualized rate under “2/10 Net 30” is therefore $(1.02)^{365/20} - 1 = 43.5\%$ if payment is made on the due date, and $(1.02)^{365/(20+30)} - 1 = 15.6\%$ once the delay is included.

To study these issues, we propose a conceptual perspective that distinguishes between two mechanisms underlying late payment behavior. Under a *flexibility mechanism*, financially healthy firms may delay payments to accommodate short-term liquidity fluctuations, using trade credit as a working-capital buffer. Under a *distress mechanism*, financially constrained firms delay payments because of operational pressure or liquidity shortages. These mechanisms generate distinct predictions regarding the drivers, persistence, and operational consequences of delayed payments.

Building on this framework, we develop three hypotheses. Because firms simultaneously receive payments from downstream customers and make payments to upstream suppliers, liquidity pressure can propagate along supply chains. When downstream customers delay payments, focal firms may face working-capital pressure and respond by postponing payments to their own suppliers. We refer to this downstream-generated liquidity pressure as *downstream cost-shifting needs*, which captures firms' exposure to delayed payments from downstream customers and the resulting pressure to shift financing costs upstream to suppliers. Accordingly, firms with greater downstream cost-shifting needs should be more likely to delay payments to their suppliers (Hypothesis 1). Second, if payment delays serve as a discretionary liquidity-management tool under the flexibility mechanism, they should remain relatively short-lived. In contrast, distress-driven delays are more likely to become persistent and trigger adverse supplier responses, such as litigation or relationship termination (Hypothesis 2). Finally, flexibility-driven delays should allow financially healthy firms to maintain operational stability—such as stable inventory turnover—while using payment timing as a working-capital buffer. In contrast, distress-driven delays are more likely to be associated with operational pressure and inventory adjustments among financially constrained firms. At the same time, delayed payments may impose financial costs, such as forgone early-payment discounts or supplier price premiums, leading to lower profitability (Hypothesis 3).

We test these hypotheses using a unique dataset from Dun & Bradstreet (D&B) that records supplier-reported payment experiences. The data provide granular information on the timing of customer payments relative to contractual terms. Unlike most existing datasets, which observe trade credit balances only through accounting variables, these records capture realized payment behavior. This feature allows us to move beyond the traditional binary distinction between repayment and default and instead analyze payment delays as a continuous behavioral dimension.

Our empirical results support the three hypotheses. First, consistent with Hypothesis 1, firms with greater downstream cost-shifting needs—proxied by longer accounts receivable days—are significantly more likely to delay payments to their suppliers. Importantly, this association is observed primarily among financially healthy firms, suggesting that many payment delays reflect discretionary liquidity management rather than distress propagation. Second, consistent with Hypothesis 2, most payment delays are relatively short-lived, while persistent delays are much less common

and are more often associated with supplier enforcement actions such as litigation or relationship termination. Third, consistent with Hypothesis 3, delayed payments are associated with lower profitability, reflecting financial costs such as forgone discounts or supplier price premiums. The operational implications, however, differ across firms: financially healthy firms maintain relatively stable inventory turnover despite delaying payments, whereas financially distressed firms exhibit higher turnover, consistent with accelerated inventory cycling under operational pressure.

This study makes two primary contributions to the operations–finance literature on trade credit. First, we introduce a novel empirical dataset that provides direct visibility into realized trade credit payment behavior. By observing payment timing at scale, our study moves beyond the traditional binary distinction between repayment and default and instead examines payment delays as a continuous behavioral dimension. Second, we propose and empirically support a unified framework that distinguishes between flexibility-driven and distress-driven payment delays. By showing that the interpretation of late payment depends on firms’ financial conditions and the persistence of delays, our framework helps reconcile conflicting interpretations in the literature that have traditionally viewed payment delays primarily as signals of distress.

Our findings also have implications for managers and policymakers concerned with supply chain financing. For managers, the results highlight the importance of balancing liquidity flexibility against relationship stability when managing trade credit payments. Moderate payment delays can provide working-capital flexibility without disrupting operations, but persistent delays risk damaging supplier relationships and destabilizing supply chains. In addition, our results suggest that managers can better interpret customer payment delays: delays from firms with stable inventory levels are more consistent with benign flexibility-driven adjustments, whereas delays accompanied by volatile inventory patterns may signal distress-driven pressures and elevated customer risk. For policymakers, the results suggest that regulatory interventions should focus primarily on mitigating persistent and systemic payment delays rather than restricting short-term liquidity adjustments that facilitate supply chain coordination.

The remainder of this paper is structured as follows. Section 2 provides a literature review and our hypothesis development. Section 3 offers a detailed description of the data and variables used in our study. In Section 4, we present the empirical baseline results together with robustness tests, supporting Hypothesis 1. Section 5 further examines the duration of late payments and the responses from suppliers, supporting Hypothesis 2. The operational outcomes with respect to Hypothesis 3 are presented in Section 6, and Section 7 concludes.

2. Theoretical Framework and Hypothesis Development

2.1. Literature Review: Why Late Payment Matters

Trade credit is a dominant source of working capital financing and a key instrument for coordinating decisions in supply chains. A large operations–finance literature shows that trade credit can shape

inventory decisions and improve supply chain coordination by aligning operational and financing incentives (Haley and Higgins 1973, Kouvelis and Zhao 2012, Yang and Birge 2018). Yet, despite extensive work on suppliers' provision of trade credit (Rajan and Zingales 1995, Petersen and Rajan 1997) and its coordination role (Babich and Tang 2012, Chod 2017), less is known empirically about buyers' payment behavior—especially the pervasive practice of paying late rather than either paying exactly on time or defaulting. This omission is notable because payment timing, unlike trade credit terms themselves, is largely determined ex post by buyers and may provide an important but underexplored means through which firms accommodate working-capital pressure.

This gap is consequential because the dominant theoretical framing in the literature has primarily interpreted late payment through the lens of financial distress and systemic risk. In the credit contagion tradition, delays and nonpayments are modeled as the propagation of liquidity shocks along supply chains, whereby financially constrained downstream firms transmit distress to their upstream suppliers (Boissay and Gropp 2013, Jacobson and Von Schedvin 2015, Agca et al. 2022). Empirical research in corporate finance similarly emphasizes the role of trade credit during periods of financial stress, but typically lacks access to granular data on the timing and duration of customer payments (Cunat 2007, Chen et al. 2022). Within this view, late payment is largely treated as an outcome of binding financial constraints, reflecting an inability to pay rather than an intentional or bounded managerial response. In turn, this perspective offers limited guidance on how firms adjust payment timing when financial constraints are present but not binding.

A complementary but distinct perspective, however, views trade credit and payment timing as part of firms' broader liquidity management toolkit. Historically, this literature characterizes trade credit as a high-cost substitute for external financing, particularly relevant for firms facing financing frictions (Smith 1987, Biais and Gollier 1997). More recent evidence challenges this narrow interpretation by documenting that even solvent firms commonly delay payments, suggesting that payment timing may serve as an operational tool through which firms accommodate short-term working-capital fluctuations (Strom 2015, McGlade 2018, Amberg et al. 2021). Moreover, Kim and Shin (2012) interpret trade credit and payment delays as reflecting buyer market power or moral hazard controls in buyer–supplier relationships, and document a close relationship between aggregate trade credit borrowing and lending. From this perspective, late payment can function as an operational buffer, allowing firms to temporarily absorb downstream cash-flow pressure without immediately adjusting production plans, inventory policies, or prices. At the same time, this buffering role is inherently bounded, as prolonged delays risk triggering supplier enforcement or relationship deterioration.

Taken together, these two streams point to a fundamental tension in interpreting late payment behavior. Do observed payment delays primarily reflect financial distress and contagion, or do

they represent a flexible but limited response to working-capital pressure, which can be moderated by market position? Disentangling these interpretations requires moving beyond binary notions of repayment and examining how payment timing is adjusted, how long delays persist, and what operational consequences follow.

Especially for the last question, the supply chain finance stream has emphasized that financing choices shape operational performance, but the direction of the effect can be nuanced. Theory highlights that trade credit can facilitate better inventory decisions (Haley and Higgins 1973, Kouvelis and Zhao 2012, Yang and Birge 2018), while empirical work finds mixed effects, including potential efficiency gains alongside profitability erosion (Peura et al. 2017) and relationship stability considerations (Reindorp et al. 2018). Importantly, this literature suggests that the consequences of financing arrangements depend not only on their availability, but also on how they are structured and deployed within operational processes over time. Building on this insight, we posit that payment timing—while largely unexplored in prior work—may similarly shape operational outcomes, with effects that may depend on the duration and context of delays rather than on their mere occurrence.

2.2. Hypothesis Development

We analyze a focal firm positioned between downstream customers and upstream suppliers. When downstream customers pay slowly, the focal firm’s cash is tied up in receivables, increasing short-run financing pressure. We refer to this exposure as *downstream cost-shifting needs*. Empirically, we proxy downstream cost-shifting needs using accounts receivable days (ARD)—a standard working-capital measure capturing collection delay and downstream cash conversion frictions.

When facing downstream payment delays, firms must absorb the resulting working-capital pressure through some operational or financial response. In principle, firms may attempt to pass this pressure on through price adjustments, quantity reductions, inventory policy changes, or expanded external financing. However, these alternatives are often costly or slow to implement in the short run: price changes require renegotiation and may undermine competitiveness; quantity or inventory adjustments can disrupt operations and service levels; and additional borrowing may involve approval delays, signaling concerns, or covenant constraints. By contrast, payment timing to upstream suppliers is typically immediate, unilateral, and reversible, making it a particularly accessible response to short-term liquidity pressure. As a result, downstream payment delays naturally increase incentives for firms to adjust their own payment timing upstream.

As discussed in Section 2.1, the same observable behavior of paying suppliers late can arise from two distinct underlying mechanisms. Under the *flexibility mechanism*, firms with sufficient liquidity and relational capital may adjust payment timing as a bounded response to downstream working-capital pressure, using *moderate* delays to re-time cash outflows without materially disrupting

supplier relationships. Under the *distress mechanism*, late payment reflects binding liquidity constraints rather than discretionary adjustment: firms delay payments because they lack the ability to meet obligations on time, consistent with models of financial distress and contagion along supply chains (Boissay and Gropp 2013, Jacobson and Von Schedvin 2015).

Although these two mechanisms differ, both imply the same reduced-form prediction: increases in downstream cost-shifting needs are associated with more frequent late payment. Accordingly, our first hypothesis is intentionally stated at the aggregate level. Instead, the distinction between the flexibility and distress mechanisms guides our subsequent analysis, where we examine payment duration and firm financial conditions to infer the underlying mechanism.

Hypothesis 1 (Downstream cost-shifting and late payment). Firms with higher downstream cost-shifting needs are associated with higher trade credit late payment to their upstream suppliers.

Our framework implies that the *duration* of payment delays serves as a way to distinguish between the flexibility and distress mechanisms introduced above. Payment timing is not a binary outcome but exhibits substantial continuous variation in observed delay duration, and supplier responses are likely to be nonlinear in delay length. Short payment delays are often absorbed within routine relational tolerance, whereas prolonged delays are more likely to be interpreted as relational breach, triggering intensified collection efforts, tighter trade credit terms, or relationship termination (Cunat 2007, Reindorp et al. 2018). Accordingly, delay duration provides a useful empirical diagnostic for whether late payment reflects a bounded adjustment under the flexibility mechanism or a binding constraint under the distress mechanism.

Under the flexibility mechanism, firms may accommodate downstream cost-shifting needs through short-term adjustments in payment timing. Because firms operating under this mechanism retain the capacity to restore timely payment, downstream pressure is expected to be reflected primarily in *short-term* payment delays that provide temporary liquidity relief while remaining within relational tolerance. In contrast, under the distress mechanism, late payment reflects binding liquidity constraints rather than a discretionary response. In this case, prolonged payment delays arise because firms are unable to resolve liquidity shortfalls, not because delay duration is actively adjusted in response to incremental downstream pressure. As a result, long-term delays under the distress mechanism are consistent with persistent constraint rather than a bounded adjustment.

We therefore use firm financial conditions as a diagnostic lens to interpret how downstream cost-shifting needs manifest across delay durations. If the flexibility mechanism dominates, increases in downstream cost-shifting needs should be reflected primarily in *short-term* delays, particularly among financially healthier firms that retain the capacity to restore timely payment. If the distress

mechanism dominates, late payment is more likely to take the form of *persistent* delays, with delay duration largely decoupled from marginal changes in downstream cost-shifting needs.

Hypothesis 2 (Delay duration and underlying mechanism). Downstream cost-shifting needs are associated primarily with *short-term* payment delays, consistent with the flexibility mechanism, while long-term payment delays are more indicative of the distress mechanism and largely decoupled from downstream cost-shifting needs.

Finally, trade credit payment timing can affect *operational performance* through a flexibility–cost trade-off. On the benefit side, delaying payments can free short-run liquidity and relax working-capital constraints, allowing firms to sustain ongoing operational activities in the presence of downstream pressure. On the cost side, however, late payment can deteriorate operational performance through higher input costs, including forfeited early-payment discounts, supplier price premiums, and relationship frictions that raise procurement costs (Chod 2017, Yang and Birge 2018). These cost-side operational penalties are likely to intensify as payment delays become prolonged and relational enforcement escalates (Cunat 2007). Accordingly, even when late payment provides short-run liquidity relief, it is expected to lower profitability.

Trade credit payment timing can also affect inventory-related operational outcomes, but the direction of this effect depends on the underlying mechanism. Theory highlights that trade credit can support inventory management by easing financing frictions (Haley and Higgins 1973, Kouvelis and Zhao 2012, Yang and Birge 2018). That said, such liquidity relief does not necessarily imply operational expansion; instead, it may simply allow firms to stabilize inventory policies in the face of downstream pressure. Under the flexibility mechanism, late payment helps firms avoid operational disruption, implying limited changes in inventory intensity. By contrast, under the distress mechanism, persistent payment delays reflect binding liquidity constraints that can force firms to compress operations, potentially accelerating inventory liquidation rather than improving operational efficiency. As a result, observed changes in inventory turnover may reflect constrained adjustment under distress rather than flexibility-driven improvement.

Hypothesis 3 (Operational outcomes). Late payment is associated with lower profitability. Its association with inventory turnover depends on the underlying mechanism: under the flexibility mechanism, late payment is associated with stable inventory turnover, whereas under the distress mechanism, it is associated with higher turnover, consistent with operational compression.

3. Data and Variables

In Section 3.1, we introduce our trade credit payment data from Dun & Bradstreet (D&B), which provides one of the most extensive trade financing programs.² This dataset allows us to observe real-

² <https://www.dnb.com/products/finance-credit-risk/global-trade-exchange-program.html>

ized payment timing and delay duration, which are central to our conceptual distinction between bounded flexibility and distress-driven delays. We merge the D&B, Compustat, and FactSet Revere databases to construct firm-year-level financial variables. Subsequently, we describe the construction and interpretation of key variables, explicitly linking them to the theoretical mechanisms developed in Section 2. This includes the dependent variables of trade credit late payments, discussed in Section 3.2, and the measure of downstream cost-shifting needs (ARD), related controls, and operational outcomes in Section 3.3.

3.1. Trade Credit Payment Dataset

We use the late payment database from the D&B Global Trade Exchange Program. D&B collects and updates trade terms, invoices, and customer payments from companies participating in the program on a daily basis. The quality and reliability of the data are supported by D&B’s “Give to Get” arrangement with the participating supplier firms. Although firms voluntarily join the program, the agreement stipulates that, once they join, they must disclose all trade information for all downstream customers to D&B. In return, D&B provides firm-level trade credit information to participating firms, thereby revealing the payment behavior of customer firms at an aggregate level. We note that D&B is legally prohibited from publishing trade credit information on any specific relationship between suppliers and customers. Therefore, firms are not subject to concerns regarding the leakage of relationship-specific information through the program. D&B then aggregated variables at the firm level at a monthly frequency, including the late payment experience, in which an experience refers to an account with D&B. The dataset encompasses trade credit information of more than 7,000 unique U.S.-headquartered public firms over 2005–2015, corresponding to roughly 70% of the Compustat universe in any given year (see Online Appendix OA1 for details).

The D&B database also categorizes late payment experiences by periods, including late payments for less than 30/60/90/120/180 and more than 180 days. This granularity allows us to observe the delay-duration patterns of late payment behavior. Consequently, the aggregated trade credit payment information collected by D&B offers more granular information than the common alternative trade credit data, which are accounts receivable and payable in a firm’s financial statements.

Note that although the raw data for each firm i are at the monthly level, our analysis is conducted at the yearly level after merging D&B, Compustat, and FactSet Revere, as we focus on the late payment information in D&B data. Therefore, the time dimension, represented by t , is aggregated at the yearly level for our main analysis in this paper. In Appendix A, we provide a descriptive summary of D&B variables, such as an example record that illustrates the granular structure of the data (Table A1), and the variables used in this study, including the total number of payment experiences in the year ($total_exp_{it}$), the number of suppliers with discount terms, and the number of late payment experiences in that year ($late_exp_{it}$) (Table A2).

3.2. Late Payment Experience

We define our dependent variable – late payment experience – as:

$$LPE_{i,t} = \frac{late_exp_{i,t}}{total_exp_{i,t}}, \quad (1)$$

which estimates the average likelihood that a firm delays a trade credit payment to any of its suppliers, regardless of the transaction value.

Next, we merge the D&B trade credit payment data with Compustat financial variables, and drop firm observations in the agriculture (i.e., SIC from 0111 to 0999), finance (SIC from 6000 to 6999), and public sector (SIC from 9000 to 9999) industries, which are less relevant for studying trade credit. We use firm-year observations by taking the annual averages of LPE , matching the independent variables of firms' fundamentals from the annual financial reports. This annual resolution is sufficient, as we will show in the empirical results that the practice of trade credit late payment varies greatly among firms but changes slowly over time. We also conduct a set of robustness tests in Online Appendix OA2 using quarterly data, and the results are consistent.

In Figure 1, we show the time trend of LPE_{it} from 2005 to 2015. The variable LPE declines steadily from 2005 through 2010 and stabilizes thereafter at an average level of approximately 24%. It is important to distinguish between trade credit late payment (i.e., LPE_{it}) and credit defaults, which peak around the 2008 financial crisis. Defaults reflect a complete failure to meet obligations, whereas late payments capture delayed—but ultimately fulfilled—transactions. This distinction motivates our focus on payment timing as an operational response, rather than as an indicator of outright financial failure.

We further decompose late payments by duration, distinguishing between short-term overdue trade credit (1-60 days late) and long-term overdue trade credit (more than 60 days late) to construct the corresponding short-/long-term LPE variables. As demonstrated in Figure 1, the decrease in LPE over the sample period is largely attributed to the decline in short-term LPE , while long-term LPE remains stable. Meanwhile, long-term LPE is substantially less prevalent in levels: 77.8% of late payment experiences ($late_exp_{i,t}$) are short-term. This distributional pattern suggests that delayed payment is most commonly observed within a limited and bounded window, rather than through prolonged delinquency.

3.3. Variable Definitions

Next, we define our explanatory variable. We measure a firm's *downstream cost-shifting needs* by using its accounts receivable days (ARD_{it}), which is calculated by taking the ratio of the current year's accounts receivable to sales and then multiplying it by 365. ARD_{it} reflects the extent to which a firm's cash inflows are delayed by downstream customers and therefore the exposure to downstream working-capital pressure rather than default per se. Prior research shows that payment

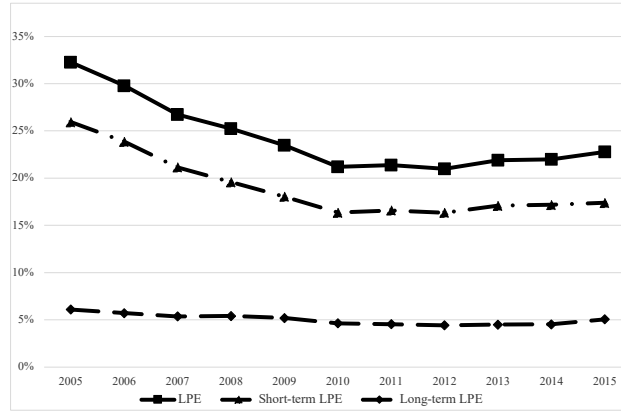


Figure 1 Late Payment Experience over Time.

problems and financial stress can propagate along supply chains, with downstream customers' payment behavior affecting upstream firms' financial conditions (Jacobson and Von Schedvin 2015). In this context, trade credit late payment can be viewed as a re-timing of cash flows along the supply chain, whereby downstream delays increase short-run financing pressure on the focal firm. Consistent with this interpretation, a higher level of ARD_{it} implies that a larger share of operating cash is tied up in outstanding claims on customers, increasing the firm's exposure to short-term liquidity frictions (Boissay and Gropp 2013). Longer receivable days therefore do not necessarily indicate financial distress, but they do increase the firm's need to accommodate delayed cash inflows. As developed in Section 2, this exposure creates incentives for firms to adjust their own payment timing upstream, making ARD_{it} a reasonable proxy for downstream cost-shifting needs in our empirical analysis.

We include a series of control variables capturing firms' access to external financing, financing priorities, and competitive conditions in the supply chain. The first set of controls comprises variables that capture access to external financing channels, particularly bank finance. Prior work shows that suppliers may have informational and monitoring advantages over banks in providing short-term financing (Petersen and Rajan 1997, Biais and Gollier 1997, Kouvelis and Zhao 2012, Chod 2017, Yang and Birge 2018), making it important to account for firms' alternative financing capacity. We control for firm size ($Size_{it}$), measured by total assets, as larger firms generally have better access to external capital markets and greater capacity to meet payment obligations on time. Next, we control for the cost of borrowing (COB_{it}), defined as the total interest paid divided by the sum of long-term and short-term debt, to capture the relative costliness of bank financing; higher borrowing costs may increase reliance on trade credit timing as a short-run liquidity adjustment. We also control for a firm's failure probability (FP_{it}). In Campbell et al. (2008), a financial "failure" includes bankruptcies (Chapter 7 or 11), financially driven delistings, and D-rated firms. FP_{it}

is computed as $1/(1 + \exp(-z))$, where z is a linear combination of predictors such as net income, liabilities, and cash, divided by assets. Higher FP_{it} reflects weaker financial health and limited access to external financing, conditions under which payment delays may arise from constraints rather than discretion.

The second group of controls reflects financing priorities consistent with pecking-order considerations (Myers and Majluf 1984). Firms may prefer to preserve internal liquidity and maintain bank relationships before meeting supplier obligations. Thus, we first include cash holdings using the cash ratio ($Cash_{it}$), defined as cash and marketable securities scaled by total assets, as firms with greater internal liquidity are better positioned to make timely supplier payments. We then include the bank debt ratio (BDR_{it}), measured as bank loans scaled by cost of goods sold (COGS), which proxies for firms' reliance on bank financing and their incentives to prioritize bank repayment.

Another set of controls captures competitive conditions within supply chains. To account for horizontal competition, we include horizontal accounts receivable days ($HARD_{it}$), defined as the leave-one-out average receivable days among firms (excluding the focal firm) in the same three-digit SIC industry, which captures competitive spillovers in downstream credit provision (Peura et al. 2017, Chod et al. 2019). To capture vertical bargaining power, we control for the average size of firm i 's downstream customers ($CSize_{it}$), constructed using supplier–customer links from the FactSet Revere database. Larger downstream customers may exert greater bargaining power, increasing the focal firm's exposure to delayed cash inflows and shaping its working-capital adjustments. Finally, we control for discount terms using the ratio of discounted trade credit experiences ($DiscExp_{it}$), following Petersen and Rajan (1997), to account for contractual incentives that may discipline payment timing.

Next, we define the outcome variables used in Hypothesis 3. Theoretical work suggests that trade credit can enhance operational outcomes. Specifically, Haley and Higgins (1973) and Kouvelis and Zhao (2012) argue that trade credit can support inventory-related decisions by decoupling inventory financing from immediate cash outflows. However, this literature does not imply that delayed payment mechanically improves operational efficiency; rather, it highlights the potential for payment flexibility to stabilize operations in the presence of financing frictions. To capture inventory-related operational outcomes, we use inventory turnover, defined as the ratio of COGS to the average inventory over the year (Alan et al. 2014). Inventory turnover is a standard operations metric, but its interpretation is context-dependent: higher turnover may reflect smoother flow and efficient replenishment, or, alternatively, accelerated inventory depletion under constraint. Relatedly, Hsu and Wu (2023) document that firms hold and adjust inventory for financial as well as operational motives. Accordingly, in our analysis inventory turnover serves as a diagnostic outcome whose interpretation depends on the underlying payment-delay mechanism.

To capture the operational costs associated with late payment, we use the ratio of sales to COGS, which we refer to as *Profitability*. This measure reflects the firm’s ability to preserve operating markups in the presence of higher input costs, forgone early-payment discounts, supplier price premiums, or relationship frictions. Consistent with our framework, we treat profitability as an operational outcome that captures the cost side of payment flexibility, rather than as a broad measure of financial performance.

Finally, some variables are highly skewed and are affected by extreme values. In the following empirical tests, similar to many papers in the finance literature, we take logarithms of the financial-related variables (i.e., $Size_{it}$, $CSize_{it}$). When a financial variable can be zero, we add one prior to taking the logarithm to keep the observations (i.e., ARD_{it} , COB_{it} , $FP_{i,t}$, $Cash_{it}$, $BDR_{i,t}$, and $HARD_{it}$). Moreover, to avoid the effects of extreme values, all variables are winsorized at 1% and 99%. Table 1 provides a complete list of variable definitions and the summary statistics of variables after the log transformation and winsorization. Finally, we also checked the pairwise correlations among the key variables and found no strong correlation.

Table 1 Variable Definitions and Summary Statistics

Variable	Definition	N	Mean	SD
$LPE_{i,t}$	Late payment experience, defined as $late_exp_{i,t}/total_exp_{i,t}$.	8,686	0.298	0.117
$LPE_short_{i,t}$	Short-term late payment experience (≤ 60 days past due).	8,686	0.233	0.097
$LPE_long_{i,t}$	Long-term late payment experience (> 60 days past due).	8,686	0.063	0.041
$ARD_{i,t}$	Log of (accounts receivable days + 1), where accounts receivable days is accounts receivable divided by sales, multiplied by 365.	8,686	3.926	0.691
$Size_{i,t}$	Log of total assets.	8,686	7.051	2.064
$COB_{i,t}$	Log of (cost of borrowing + 1), where cost of borrowing is total interest paid divided by the sum of long-term and short-term debt.	8,686	0.090	0.174
$FP_{i,t}$	Log of (failure probability + 1), following Campbell et al. (2008) .	8,686	0.008	0.036
$Cash_{i,t}$	Log of (cash ratio + 1), where cash ratio is cash and marketable securities divided by total assets.	8,686	0.143	0.144
$BDR_{i,t}$	Log of (bank debt ratio + 1), where bank debt ratio is current liability (a proxy for bank loans) divided by COGS.	8,686	0.080	0.165
$HARD_{i,t}$	Log of (horizontal ARD + 1), where horizontal ARD is the average accounts receivable days of competitors (excl. the focal firm) in the same industry.	8,686	4.034	0.479
$CSize_{i,t}$	Average log of total assets of the firm’s downstream customers.	8,686	9.174	1.771
$Disc.Exp_{i,t}$	The proportion of suppliers with discount terms divided by the total number of suppliers with trade credit.	8,686	0.155	0.073
$Inv.Turn_{i,t}$	The ratio of COGS to the average inventory of the year.	7,334	17.604	44.581
$Profitability_{i,t}$	The ratio of sales to COGS.	8,680	2.051	1.668

4. Downstream Cost-shifting Needs

In this section, we examine Hypothesis 1. We focus on establishing the reduced-form relationship between downstream payment pressure and upstream payment delays, without attributing it to a specific mechanism. Section 4.1 presents the baseline results. We then assess robustness using an instrumental-variable approach and exogenous natural disaster shocks affecting downstream

customers in Section 4.2. The interpretation of this association—through payment duration and firm financial conditions, as articulated in Hypothesis 2—is examined in Section 5.

4.1. Empirical Results for Hypothesis 1

We use the following specification to test Hypothesis 1, i.e., whether a firm with higher downstream cost-shifting needs is associated with more late payments to its suppliers:

$$\begin{aligned} LPE_{i,t} = & \beta_0 + \beta_1 ARD_{i,t-1} + \beta_2 Size_{i,t-1} + \beta_3 COB_{i,t-1} + \beta_4 FP_{i,t-1} + \\ & \beta_5 Cash_{i,t-1} + \beta_6 BDR_{i,t-1} + \beta_7 HARD_{i,t-1} + \beta_8 CSize_{i,t-1} + \\ & \beta_9 Disc_Exp_{i,t-1} + \text{Fixed Effects} + \epsilon_{i,t}, \end{aligned} \quad (2)$$

in which we include several fixed effects combinations for the robustness of the results, including the year fixed effects, the industry fixed effects, and the stringent industry-year joint fixed effects.

Prior literature shows that trade credit plays an important role in firms' financing structure and working-capital management (Kouvelis and Zhao 2012, Yang and Birge 2018). Accordingly, our baseline specifications use one-year-lagged explanatory variables to relate firms' exposure to downstream cost-shifting needs to subsequent payment behavior. This timing reduces concerns about simultaneity and reverse causality, while remaining consistent with our framework in which payment timing is a short-run adjustment to downstream cost-shifting needs rather than a contemporaneous reaction to realized distress. As we focus on the cross-sectional differences in firm characteristics to explain the trade credit late payment behaviors, we do not include firm fixed effects; instead we cluster robust standard errors at the firm level.³

Table 2 presents the regression results. We first run the model in Equation (2) without any fixed effects (Model 1), then with only yearly fixed effects (Model 2), with separate yearly and industry fixed effects (Model 3), and finally with year-industry joint fixed effects (Model 4). We first note that the statistical significance levels are consistent controlling for different combinations of the year and the industry fixed effects. Specifically, all specifications from Models 1 to 4 show that a firm's accounts receivable days (*ARD*) are positively and significantly associated with its late payment experiences to its suppliers.

While the positive *ARD*–*LPE* association in the pooled sample supports Hypothesis 1, aggregate correlations alone cannot distinguish whether late payment reflects discretionary flexibility or binding distress. Accordingly, in Section 5, we use payment duration and firm financial conditions as diagnostic tools to infer the underlying mechanism and test Hypothesis 2.

Next, we assess the economic significance of the key explanatory variables using the most stringent specification (Model 4); see Table 3. Column (1) reports the change in *LPE* associated with a

³ Online Appendix OA2 discusses this choice in detail and reports a quarterly specification that includes firm fixed effects; the results for Hypotheses 1 and 2 are consistent.

Table 2 Regression Results for Hypothesis 1

Variable	(1)	(2)	(3)	(4)
<i>ARD</i>	0.009*** (2.771)	0.009*** (2.765)	0.008** (2.511)	0.008** (2.469)
<i>Size</i>	0.014*** (11.360)	0.014*** (11.230)	0.016*** (12.497)	0.016*** (12.493)
<i>COB</i>	0.006 (0.667)	0.008 (0.857)	0.008 (0.891)	0.008 (0.874)
<i>FP</i>	0.044 (1.315)	0.108*** (2.867)	0.117*** (3.256)	0.117*** (3.175)
<i>Cash</i>	-0.098*** (-6.692)	-0.098*** (-6.644)	-0.106*** (-6.910)	-0.108*** (-7.004)
<i>BDR</i>	-0.038*** (-3.760)	-0.039*** (-3.854)	-0.026*** (-2.676)	-0.026*** (-2.670)
<i>HARD</i>	0.010* (1.706)	0.009 (1.626)	0.003 (0.539)	0.004 (0.635)
<i>CSize</i>	-0.002** (-2.176)	-0.003** (-2.175)	-0.001 (-1.271)	-0.002 (-1.336)
<i>Disc_Exp</i>	0.698*** (21.757)	0.689*** (18.164)	0.578*** (14.025)	0.579*** (13.827)
Fixed Effects	None	Year	Year+Ind	Year×Ind
Observations	8,686	8,686	8,686	8,686
R-squared	0.269	0.274	0.310	0.320

Note. The dependent variable is *LPE*. Models 1–4 report OLS estimates on the pooled sample with progressively richer fixed effects. Robust standard errors are clustered by firm. t-statistics are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1 using two-tailed tests.

one-standard-deviation increase in each variable, and Column (2) reports the corresponding standardized effect, computed relative to the sample standard deviation of *LPE* (0.117). Accounts receivable days (*ARD*), interpreted as exposure to delayed downstream cash inflows, exhibit a non-trivial economic magnitude. A one-standard-deviation increase in *ARD* is associated with a 4.72% increase in *LPE* (in standardized terms), indicating that downstream payment exposure is comparable in magnitude to other core firm characteristics. For comparison, a one-standard-deviation increase in firm size (*Size*) is associated with a 28.23% increase in *LPE*. This is consistent with prior literature that payment delay can be viewed as an exercise of market power or moral hazard control over suppliers by dominant customers (Kim and Shin 2012) and prior evidence that larger firms exhibit more delayed payment behavior in trade credit relationships (Quick 2013). By contrast, liquidity availability is associated with less late payment: a one-standard-deviation increase in cash holdings (*Cash*) is associated with a 13.29% decrease in *LPE*. Contract terms are also quantitatively meaningful, as a one-standard-deviation increase in the prevalence of discount terms (*Disc_Exp*) is associated with a 36.13% increase in *LPE*. These comparisons indicate that *ARD* is economically meaningful alongside standard financial and contractual determinants of payment timing.

Table 3 Economic Significance on LPE

Variable	<i>LPE</i> (raw value)	<i>LPE</i> (relative ratio)
<i>ARD</i>	0.55%	4.72%
<i>Size</i>	3.30%	28.23%
<i>COB</i>	0.14%	1.19%
<i>FP</i>	0.42%	3.60%
<i>Cash</i>	-1.56%	-13.29%
<i>BDR</i>	-0.43%	-3.67%
<i>HARD</i>	0.19%	1.64%
<i>CSize</i>	-0.35%	-3.03%
<i>Disc_Exp</i>	4.23%	36.13%

Note. We calculate the economic significance (raw value columns) by taking the product of the coefficients and the respective standard deviations, whereas the relative ratio is calculated by taking the *LPE* (raw value columns) divided by the standard deviations of *LPE*.

Moreover, to provide additional intuition for why even moderate payment delays can be economically meaningful, we offer a simple calculation that illustrates the potential gross liquidity benefit of short-term payment delays.⁴ Consider a firm with an annual cost of debt of approximately 6% (about 0.5% per month). By delaying a \$10 million payment for 30 days, the firm creates an interest-free liquidity buffer that saves \$50,000 in monthly interest expenses. While this amount may appear small relative to total revenue, it is non-trivial: for a firm with a 5% net profit margin, this saving is equivalent to the profit generated by \$1 million in additional sales.

4.2. Robustness Tests

Next, we conduct a series of complementary analyses designed to address identification concerns, alternative explanations, and model dependence. We first implement an instrumental variable (IV) strategy that isolates plausibly exogenous variation in downstream payment exposure. We then exploit exogenous natural disaster shocks to customers as an additional source of variation that sharpens the interpretation of *ARD*.⁵

For the IV analysis, we instrument *ARD* using customer dispersion (*Cdispersion*), defined as $1/(1 + N_{customers})$, where $N_{customers}$ is the number of downstream customers identified from the FactSet Revere database. According to Wooldridge (2012), valid instruments must satisfy the following two conditions: they must be “uncorrelated with the error (exogeneity) and correlated with the endogenous explanatory variable (relevance)”, in which the error term refers to $\epsilon_{i,t}$ in Equation (2). A high *Cdispersion* value means that the focal firm serves a small number of customers. As

⁴ This illustration isolates the gross financing benefit of deferring payment. Our profitability analysis in Section 6.1, instead, captures the net effect after accounting for forgone early-payment discounts, supplier price premiums, and relationship frictions, which can offset the financing benefit, resulting in a negative association between late payment and profitability.

⁵ Online Appendix OA3 reports three additional checks: controlling for industry peer payment norms, a crisis-period (2007–2009) split, and Fama–MacBeth two-step estimation.

Lee et al. (2018) show, a more concentrated customer base also leads to lower trade credit provision, thereby lowering accounts receivable days. Regarding the exogeneity condition, *Cdispersion* is constructed on the basis of the focal firm's number of customers, whereas the dependent variable is based on its suppliers. Customer dispersion is less likely to affect supplier-side payment delays except through the focal firm's receivables exposure.

In Models 1 and 2 of Table 4, we report the first and second stages of the IV regressions, respectively. The coefficient of *Cdispersion* in Model 1 is significant at the 1% level, implying the relevance condition, and the Kleibergen-Paap F statistic of 45.46 indicates a strong instrument. The second-stage results remain consistent with Hypothesis 1.⁶

To further probe identification concerns, we exploit exogenous shocks to downstream payment exposure arising from major natural disasters. This analysis leverages plausibly idiosyncratic disruptions to customers' operations and liquidity that are less related to the focal firm's own payment behavior, providing an additional source of variation in downstream cost-shifting needs. Following Agca et al. (2022), we obtain customer establishment locations from the Environmental Protection Agency and identify major natural disasters using data from the Federal Emergency Management Agency. We then construct a firm-level measure of downstream disaster exposure as follows:

$$Cdisaster_{i,t} = \frac{\sum_{j=1}^{N_{i,t}} D_{j,t}}{N_{i,t}}, \quad (3)$$

where $D_{j,t}$ equals one if at least one facility of customer j is affected by a major natural disaster in year t , and zero otherwise, and $N_{i,t}$ is the total number of customers of firm i in year t . This measure captures the proportion of a firm's customers affected by large, localized disasters that are known to have short-run effects on productivity, sales, and liquidity (Barrot and Sauvagnat 2016).

We incorporate this shock into Equation (2) by including $Cdisaster_{i,t-1}$ and its interaction with accounts receivable days, $ARD_{i,t-1} \times Cdisaster_{i,t-1}$, and re-estimate the baseline specifications. The sample size is reduced because constructing the disaster exposure measure requires identifying customer facility locations. As reported in Table 4, the interaction term is positive and statistically significant regardless of whether disasters are measured at the county (Model 3) or state level (Model 4). For example, in the county-level specification (Model 3), the coefficient on $ARD \times Cdisaster$ is 0.021, implying that for each one-unit increase in downstream disaster exposure, the

⁶ Relative to the OLS estimates in Table 2, the IV coefficient is larger in magnitude, which may reflect the fact that IV identifies a local average treatment effect (LATE) for firms whose receivable exposure is most responsive to customer dispersion. When treatment effects are heterogeneous, IV estimates capture the effect for a specific subpopulation influenced by the instrument, whereas OLS estimates average effects across all firms (Angrist 2004, Ishimaru 2024). Large differences between LATE and ATE are not necessarily unusual. Consistent with this, Jiang (2017) documents that in approximately 80% of published studies, IV estimates exceed uninstrumented estimates, often by substantial margins.

marginal effect of a one-unit increase in ARD on LPE rises by 0.021. This natural disaster analysis supports the interpretation that the baseline ARD–LPE association reflects a systematic response to downstream payment exposure rather than reverse causality or omitted firm-level factors.

Table 4 Robustness: Instrumental Variable and Natural Disaster Analysis

Variable	IV: Customer Dispersion		Natural Disaster Shock	
	1st Stage: <i>ARD</i> (1)	2nd Stage: <i>LPE</i> (2)	County: <i>LPE</i> (3)	State: <i>LPE</i> (4)
<i>Cdispersion</i>	-0.642*** (-6.742)			
<i>ARD</i>		0.075*** (2.894)	0.012** (2.400)	0.011** (2.200)
<i>Cdisaster</i>			-0.106** (-2.304)	-0.094*** (-3.133)
<i>ARD</i> × <i>Cdisaster</i>			0.021* (1.909)	0.019** (2.375)
<i>Size</i>	-0.011 (-1.406)	0.016*** (11.564)	0.015*** (7.500)	0.015*** (7.500)
<i>COB</i>	-0.075 (-1.085)	0.012 (1.190)	-0.001 (-0.083)	-0.001 (-0.091)
<i>FP</i>	-0.578* (-1.664)	0.154*** (3.180)	0.096* (1.959)	0.095* (1.939)
<i>Cash</i>	-0.956*** (-6.626)	-0.047* (-1.658)	-0.118*** (-6.556)	-0.118*** (-6.556)
<i>BDR</i>	0.396*** (3.448)	-0.052*** (-3.541)	-0.024* (-1.714)	-0.024* (-1.714)
<i>HARD</i>	0.503*** (11.013)	-0.031** (-2.090)	0.003 (0.375)	0.003 (0.375)
<i>CSize</i>	0.006 (0.850)	-0.002* (-1.760)	-0.003 (-1.500)	-0.003 (-1.500)
<i>Disc.Exp</i>	-0.007 (-0.035)	0.543*** (11.910)	0.631*** (11.906)	0.630*** (11.887)
Fixed Effects	Year × Ind	Year × Ind	Year × Ind	Year × Ind
Observations	8,686	8,686	5,150	5,150
R-squared	0.258		0.325	0.325
Kleibergen-Paap F		45.46		

Note. The dependent variable is *LPE*. Models 1 and 2 report two-stage least squares (2SLS) estimates using $Cdispersion = 1/(1 + N_{customers})$ as the instrument for *ARD*, where $N_{customers}$ is the number of downstream customers. The Kleibergen-Paap F-statistic is 45.46, exceeding the conventional Stock-Yogo weak instrument critical value of 10. Models 3 and 4 examine how county- and state-level natural-disaster exposure among customers moderates the association between *ARD* and late payments. All models include Year × Industry fixed effects. Robust standard errors are clustered by firm. t-statistics in parentheses. *** p<0.01, ** p<0.05, * p<0.1 using two-tailed tests.

5. Delay Duration and Underlying Mechanism

Having established the aggregate association between downstream cost-shifting needs and late payment in Section 4.1, we now turn to the mechanism-diagnostic analysis predicted by Hypothesis 2. Section 5.1 decomposes delays into short-term and long-term components and uses the Altman Z-score to interpret whether observed delays are more consistent with the flexibility-driven or

distress-driven mechanism. Section 5.2 then examines supplier responses and shows that long-term delays—but not short-term delays—are significantly associated with a higher risk of legal action, consistent with the relational tolerance boundary that motivates our duration decomposition.

5.1. Empirical Results for Hypothesis 2

Before decomposing delays by duration, we first revisit the aggregate ARD–LPE association to assess whether it is more consistent with the flexibility mechanism or the distress mechanism introduced in Section 2.2. Table 5 splits the pooled sample by the year-specific cross-sectional median of lagged Altman Z-score⁷ and re-estimates Equation (2). The *ARD* coefficient is positive and significant for financially healthy firms ($\beta = 0.013$ with $p < 0.01$ in Model 1) but smaller and statistically insignificant for financially distressed firms ($\beta = 0.005$ with $p > 0.1$ in Model 2). This asymmetry suggests that the aggregate ARD–LPE association is concentrated primarily among financially healthy firms, consistent with the flexibility mechanism, while the distress mechanism plays a more limited role in the pooled baseline.

Table 5 Mechanism Diagnostic: Heterogeneity Analysis by Altman Z-score

Variable	High Z (1)	Low Z (2)
<i>ARD</i>	0.013*** (2.838)	0.005 (1.290)
<i>Size</i>	0.018*** (10.486)	0.016*** (9.342)
<i>COB</i>	-0.003 (-0.312)	0.030** (2.084)
<i>FP</i>	-0.066 (-0.892)	0.119*** (2.840)
<i>Cash</i>	-0.114*** (-5.598)	-0.097*** (-4.713)
<i>BDR</i>	-0.036*** (-3.230)	-0.019 (-1.416)
<i>HARD</i>	0.003 (0.398)	0.004 (0.520)
<i>CSize</i>	-0.001 (-0.549)	-0.001 (-0.783)
<i>Disc.Exp</i>	0.534*** (8.515)	0.545*** (9.951)
Fixed Effects	Year×Ind	Year×Ind
Observations	3,788	4,789
R-squared	0.357	0.320

Note. The dependent variable is *LPE*. The sample is split by the year-specific cross-sectional median of lagged Altman Z-score. Model 1 includes financially healthy firms (High Z), and Model 2 includes financially distressed firms (Low Z). All models use OLS estimation with Year × Industry fixed effects. Robust standard errors are clustered by firm. t-statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ using two-tailed tests.

⁷ The Altman Z-score (Altman 1968) is a composite measure of financial distress risk based on five accounting ratios (working capital/assets, retained earnings/assets, EBIT/assets, market equity/total liabilities, and sales/assets). We obtain the pre-calculated Z-score directly from Compustat. Higher values indicate stronger financial health.

We next examine whether this pattern extends to the *duration* of payment delays. As discussed in Section 2.2, both mechanisms predict a positive aggregate association between *ARD* and *LPE*. However, they differ in their implications for delay duration. The flexibility mechanism implies that firms accommodate downstream cost-shifting needs within a bounded relational tolerance window, leading primarily to short-term delays. In contrast, under the distress mechanism, persistent delays reflect binding financial constraints and are less sensitive to marginal variation in downstream cost-shifting needs. Duration therefore provides a diagnostic lens to distinguish between the two.

We decompose total late payment experience into short-term and long-term components:

$$LPE_short_{i,t} = \frac{late_30d_{i,t} + late_60d_{i,t}}{total_exp_{i,t}}, \text{ and} \quad (4)$$

$$LPE_long_{i,t} = \frac{late_90d_{i,t} + late_120d_{i,t} + late_180d_{i,t} + late_over180d_{i,t} + late_negative_{i,t}}{total_exp_{i,t}}. \quad (5)$$

The 60-day threshold is motivated by industry practice rather than an arbitrary cutoff. In credit management practice, accounts that become 60–90 days past due typically enter the escalation stage beyond routine collections (National Association of Credit Management 2024). Consistent with this institutional boundary, the D&B PAYDEX scoring convention shows a significant deterioration once payments move from roughly 60 to 90 days past due (Dun & Bradstreet 2026). These benchmarks support 60 days as a practically meaningful threshold separating routine flexibility from more material consequences.

Replacing the dependent variable in Equation (2) with *LPE_short* and *LPE_long*, Table 6 reports the duration results. Models 1 and 2 show that *ARD* is positively and significantly associated with *LPE_short* ($\beta = 0.006$, $p < 0.05$) but not with *LPE_long* ($\beta = 0.001$, $p > 0.1$). When we alternatively use a 90-day cutoff (Models 3–4), the pattern remains: the *ARD* coefficient drops from 0.008 ($p < 0.05$) for $LPE_{\leq 90}$ to 0 (statistically insignificant) for $LPE_{>90}$. These results suggest that downstream cost-shifting needs are associated primarily with short-term payment adjustments rather than persistent delays.

Beyond *ARD*, the coefficient magnitudes on key financial variables decline sharply when the dependent variable shifts from short-term to long-term delays. In Table 6, cash holdings (*Cash*) and bank debt ratio (*BDR*) exhibit materially stronger associations with *LPE_short* than with *LPE_long*. While these variables are economically and statistically meaningful predictors of short-term delay, their magnitudes shrink substantially—and in some cases lose statistical significance—when explaining long-term delay. To formally test whether these differences are statistically significant, we employ Seemingly Unrelated Estimation (SUEST) tests with firm-clustered standard errors. Table 7 (Panel A) shows that the coefficients on *Cash* ($\chi^2 = 44.62$, $p < 0.001$) and *BDR* ($\chi^2 = 25.56$, $p < 0.001$) differ significantly between the short-term and long-term models, and

Table 6 Analysis on Payment Duration

Variable	Pooled		Robustness		High Z		Low Z	
	(1) <i>LPE</i> _short	(2) <i>LPE</i> _long	(3) <i>LPE</i> _{≤90}	(4) <i>LPE</i> _{>90}	(5) <i>LPE</i> _short	(6) <i>LPE</i> _long	(7) <i>LPE</i> _short	(8) <i>LPE</i> _long
<i>ARD</i>	0.006** (2.194)	0.001 (1.065)	0.008** (2.532)	0.000 (0.463)	0.011*** (2.820)	0.002 (1.171)	0.004 (1.125)	0.001 (1.032)
<i>Size</i>	0.006*** (5.912)	0.009*** (20.220)	0.012*** (9.607)	0.004*** (17.300)	0.007*** (4.989)	0.011*** (16.985)	0.007*** (4.738)	0.009*** (14.577)
<i>COB</i>	0.003 (0.369)	0.004 (1.516)	0.007 (0.830)	0.001 (0.660)	-0.004 (-0.462)	0.001 (0.197)	0.019 (1.504)	0.011** (2.222)
<i>FP</i>	0.073** (2.390)	0.048*** (3.257)	0.097*** (2.890)	0.022** (2.565)	-0.089 (-1.617)	0.022 (0.614)	0.083** (2.450)	0.037** (2.191)
<i>Cash</i>	-0.095*** (-7.347)	-0.013** (-2.340)	-0.107*** (-7.496)	-0.001 (-0.342)	-0.105*** (-6.051)	-0.010 (-1.558)	-0.084*** (-4.854)	-0.012* (-1.696)
<i>BDR</i>	-0.031*** (-4.185)	0.006 (1.253)	-0.028*** (-3.243)	0.002 (1.019)	-0.034*** (-3.436)	-0.002 (-0.369)	-0.028*** (-2.991)	0.009 (1.431)
<i>HARD</i>	0.005 (1.129)	-0.001 (-0.251)	0.005 (0.906)	-0.001 (-0.482)	0.002 (0.402)	0.001 (0.462)	0.007 (1.111)	-0.002 (-0.798)
<i>CSize</i>	-0.001 (-0.614)	-0.001** (-2.134)	-0.001 (-1.036)	-0.000 (-1.547)	-0.000 (-0.047)	-0.001 (-1.153)	-0.000 (-0.331)	-0.001 (-1.224)
<i>Disc_Exp</i>	0.499*** (13.630)	0.059*** (4.116)	0.586*** (15.312)	0.011 (1.523)	0.499*** (9.393)	0.043** (2.209)	0.505*** (11.325)	0.062*** (3.413)
Fixed Effects	Year×Ind	Year×Ind	Year×Ind	Year×Ind	Year×Ind	Year×Ind	Year×Ind	Year×Ind
Observations	8,686	8,686	8,686	8,686	3,788	3,788	4,789	4,789
R-squared	0.386	0.298	0.412	0.204	0.335	0.342	0.332	0.223

Note. The dependent variables are LPE-related variables. Models 1–2 report pooled OLS with Year × Industry fixed effects, using 60 days as the cutoff. Models 3–4 use 90 days as the alternative cutoff. Models 5–8 split the sample by the year-specific cross-sectional median of lagged Altman Z-score, with High Z representing financially healthy firms and Low Z representing financially distressed firms. Robust standard errors are clustered by firm. t-statistics are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1 using two-tailed tests.

ARD exhibits a marginally significant difference ($\chi^2 = 3.20$, $p = 0.074$). This systematic attenuation of coefficient magnitudes, together with the SUEST results, suggests that short-term delays are more closely associated with liquidity conditions and downstream cost-shifting needs, whereas long-term delays are substantially less sensitive to these factors—consistent with a distinction between bounded payment flexibility and distress-related delay.

To identify which firms drive the short-term flexibility pattern, we again split the sample by lagged Altman Z-score (Models 5–8 of Table 6). The *ARD*–*LPE*_short association is positive and statistically significant for financially healthy firms ($\beta = 0.011$ with $p < 0.01$ in Model 5), but statistically insignificant for financially distressed firms ($\beta = 0.004$ with $p > 0.1$ in Model 7). Long-term delays remain statistically insignificant across both subsamples. The SUEST results by financial health (Panels B and C of Table 7) reinforce this finding. For healthy firms (Panel B), the short-term and long-term coefficients differ significantly for *ARD* ($\chi^2 = 6.15$, $p = 0.013$) and key variables including *Cash*, *BDR*, and *FP*, indicating that financially healthy firms differentiate delay duration in response to downstream cost-shifting needs and liquidity conditions. In contrast, for distressed firms (Panel C), the short-term and long-term coefficients on *ARD* do not differ

Table 7 SUEST Tests for Coefficient Differences: Short-Term vs. Long-Term Delays

Variable	Coef (ST)	Coef (LT)	χ^2	p-value	Significant Difference
<i>Panel A: Pooled Sample</i>					
<i>Cash</i>	-0.095***	-0.013**	44.62	<0.001	Yes
<i>BDR</i>	-0.031***	+0.006	25.56	<0.001	Yes
<i>FP</i>	0.073**	0.048***	0.66	0.416	No
<i>ARD</i>	0.006**	0.001	3.20	0.074	Marginal
<i>Panel B: High Z — Financially Healthy</i>					
<i>Cash</i>	-0.105***	-0.010	33.82	<0.001	Yes
<i>BDR</i>	-0.034***	-0.002	8.64	0.003	Yes
<i>FP</i>	-0.089	+0.022	3.94	0.047	Yes
<i>ARD</i>	0.011***	0.002	6.15	0.013	Yes
<i>Panel C: Low Z — Financially Distressed</i>					
<i>Cash</i>	-0.084***	-0.012*	18.55	<0.001	Yes
<i>BDR</i>	-0.028***	+0.009	17.84	<0.001	Yes
<i>FP</i>	0.083**	0.037**	1.88	0.170	No
<i>ARD</i>	0.004	0.001	0.46	0.499	No

Note. This table reports Seemingly Unrelated Estimation (SUEST) tests for coefficient equality between short-term (*LPE_short*, ≤ 60 days) and long-term (*LPE_long*, > 60 days) delay regressions. Panel A uses the pooled sample; Panels B and C are split by the year-specific cross-sectional median of lagged Altman Z-score. The null hypothesis is that the coefficient is equal across the ST and LT models. “Significant Difference” indicates whether the null hypothesis of equal coefficients is rejected at the 5% level (Yes), rejected only at the 10% level (Marginal), or not rejected (No). Standard errors are clustered by firm. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ using two-tailed tests.

statistically ($\chi^2 = 0.46$, $p = 0.499$), suggesting that distressed firms lack the discretion to selectively calibrate delay duration. These results are consistent with Hypothesis 2 that downstream cost-shifting needs are associated primarily with short-term delays among financially healthy firms, consistent with bounded payment flexibility, while long-term delays are less sensitive to marginal variation in downstream cost-shifting needs and more aligned with financial distress.

5.2. Supplier Responses toward Customer Late Payment

If payment delay duration reflects a bounded flexibility mechanism, suppliers should respond differently to short-term versus long-term delays. In particular, if short-term delays fall within a relational tolerance window, they should not systematically trigger enforcement, whereas persistent delays should increase the likelihood of escalation. We therefore examine supplier-side legal and relational responses as additional evidence on the duration distinction established in Section 5.1.

To start, we collect information about corporate litigation events from the Audit Analytics Litigation database following Hutton et al. (2015) and Cen et al. (2017).⁸ For our purposes, we collect

⁸ Audit Analytics coverage starts in 2000. It reports information on lawsuits for U.S. publicly traded firms by collecting information from corporate disclosures and corporate newswires, as well as from legal disclosures, registrations, and legal opinions filed with the SEC. Audit Analytics reports details related to the specific litigation, including detailed categorizations of lawsuits, the original date of filing, and the settlement amount. The types of lawsuits include security fraud, copyright and trademark infringement, contract violation, personal injury, etc.

all firm-level contract-related legal charges as our proxy. Litigation events are rare relative to late payment experiences. The median firm-year has zero contract-related cases, the 99th percentile has one case per year, and the sample mean is 0.0641 cases per firm-year, compared to approximately 25% late payment experience per year in the D&B data. This disparity suggests that late payment is frequently tolerated and rarely escalated to formal litigation.

Next, we estimate a logistic regression of $Lawsuit_{i,t}$ on lagged LPE measures. Specifically, we define our dependent variable, $Lawsuit_{i,t}$, as a dummy variable that equals 1 if firm i has a contract-related legal charge in year t , and 0 otherwise. We present the results in Models 1-3 of Table 8. The aggregate LPE measure is not statistically associated with litigation risk. However, the duration decomposition reveals a clear asymmetry: long-term late payments are associated with significantly higher litigation risk, whereas short-term late payments are not statistically significant. This pattern indicates that enforcement risk rises primarily when payment delays extend beyond the relational tolerance window, whereas short-term delays do not systematically trigger legal response. Consistent with Yang and Birge (2018), suppliers appear more accommodating than banks in trade credit relationships, reflecting the ongoing commercial interdependence between buyers and suppliers. This evidence underscores the “co-opetition” nature of supply chains: firms are simultaneously financial counterparties and operational collaborators. Moderate delays are often absorbed within ongoing relationships, while persistent delays prompt enforcement actions.

We next examine whether late payments undermine buyer-supplier relationships more broadly. Using the FactSet Revere database, we construct a measure of terminated buyer-supplier relationships (*NumberDropped*) and estimate OLS regressions with lagged late payment-related variables as key independent variables. The results are shown in Models 4-6 of Table 8. Consistent with the litigation evidence, long-term late payments are positively and significantly associated with relationship termination, whereas short-term late payments are not. This evidence complements Ersahin et al. (2024), which links trade credit conditions to relationship stability, and extends the literature by distinguishing between the consequences of short- and long-term delay durations.

In sum, the supplier response evidence reinforces the mechanism interpretation developed in Section 5.1. Short-term delays are largely accommodated within ongoing relationships, whereas long-term delays are associated with litigation and relationship termination. This pattern supports the view that trade credit delays operate within a bounded flexibility framework rather than reflecting broad delinquency. This perspective complements the credit contagion literature (Boissay and Gropp 2013, Jacobson and Von Schedvin 2015), which emphasizes cascading distress through supply chains, by showing that among U.S. public firms, most observed payment delays are consistent with discretionary flexibility by financially viable firms rather than contagion-driven default.

Table 8 Analysis on Corporate Contract Litigation and Terminated Buyer-Supplier Relationships

Variable	<i>Lawsuit</i>			<i>NumberDropped</i>		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>LPE</i>	1.129 (1.552)			3.311*** (4.098)		
<i>LPE_long</i>		3.661** (2.177)			22.024*** (6.431)	
<i>LPE_short</i>			0.800 (0.904)			0.561 (0.680)
<i>Size</i>	0.322*** (6.012)	0.300*** (5.389)	0.337*** (6.484)	1.196*** (11.630)	1.046*** (12.080)	1.247*** (11.493)
<i>COB</i>	-0.353 (-1.019)	-0.348 (-0.990)	-0.341 (-0.991)	0.753*** (3.034)	0.706*** (2.957)	0.790*** (3.143)
<i>FP</i>	4.230*** (2.915)	4.163*** (2.861)	4.280*** (2.954)	5.970*** (5.521)	5.177*** (4.968)	6.312*** (5.748)
<i>Cash</i>	1.117** (1.994)	1.033* (1.847)	1.082* (1.938)	3.074*** (5.012)	3.024*** (4.984)	2.799*** (4.575)
<i>BDR</i>	-0.033 (-0.097)	-0.079 (-0.231)	-0.031 (-0.091)	2.778* (1.702)	2.566* (1.647)	2.717* (1.649)
<i>HARD</i>	0.344 (1.578)	0.349 (1.600)	0.344 (1.578)	-0.128 (-0.434)	-0.107 (-0.373)	-0.108 (-0.359)
<i>CSize</i>	-0.142*** (-3.279)	-0.140*** (-3.217)	-0.143*** (-3.307)	-0.289*** (-5.944)	-0.271*** (-5.770)	-0.294*** (-6.013)
<i>Disc_Exp</i>	0.543 (0.447)	1.073 (0.927)	0.775 (0.633)	2.949** (1.968)	3.592** (2.305)	4.570*** (2.881)
Year×Ind F.E.	Y	Y	Y	Y	Y	Y
Observations	8,396	8,396	8,396	8,686	8,686	8,686
(Pseudo) R-squared	0.111	0.112	0.110	0.303	0.299	0.321

Note. The dependent variable is *Lawsuit* (indicator) in Models 1–3 and *NumberDropped* (count) in Models 4–6. Models 1–3 use logistic regressions, whereas Models 4–6 use OLS regressions. Robust standard errors are clustered by firm. t-statistics are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1 using two-tailed tests.

6. Operational Outcomes

We next examine the operational outcomes of late payments as predicted by Hypothesis 3. Section 6.1 first documents the associations between late payment behavior and firms' operational outcomes, focusing on inventory turnover and profitability. Section 6.2 then provides additional evidence on the flexibility mechanism by examining firms that never delay payments, allowing us to assess whether the absence of payment flexibility is associated with different operational patterns.

6.1. Empirical Results for Hypothesis 3

To test Hypothesis 3, we estimate a specification similar to Equation (2), replacing the dependent variable by inventory turnover and a profitability-related operating outcome proxied by the sales-to-COGS ratio. Moreover, following Kesavan et al. (2010), we include simultaneous controls for profitability and inventory turnover because these two operational metrics are mechanically interrelated; without controlling for the other, an observed association could partly reflect this structural linkage rather than an independent relationship with late payment behavior. Table 9 presents the association between late payment experience and firms' operational outcomes.

Table 9 Operational Outcomes by Financial Health: Inventory Turnover and Profitability

	Inventory Turnover				Profitability			
	Pooled (1)	High Z (2)	Low Z (3)	NoDelay (4)	Pooled (5)	High Z (6)	Low Z (7)	NoDelay (8)
LPE	18.708** (2.412)	-4.892 (-0.604)	23.237** (2.212)	15.993** (2.014)	-0.587*** (-2.723)	-0.533* (-1.704)	-0.587** (-2.224)	-0.794*** (-3.491)
NoDelay				-5.080* (-1.879)				-0.391*** (-3.568)
Profitability	-3.651*** (-4.858)	-2.874*** (-4.036)	-4.925*** (-4.382)	-3.691*** (-4.882)				
Inv. Turn					-0.004*** (-4.849)	-0.004*** (-3.328)	-0.004*** (-5.061)	-0.004*** (-4.883)
Size	-0.958 (-1.616)	0.267 (0.424)	-1.026 (-1.431)	-0.946 (-1.600)	0.061*** (3.319)	0.116*** (3.975)	0.017 (0.848)	0.062*** (3.359)
COB	2.829 (0.564)	5.859 (1.076)	2.307 (0.309)	2.897 (0.577)	-0.097 (-1.171)	0.054 (0.537)	-0.234* (-1.715)	-0.092 (-1.107)
FP	28.294 (1.430)	12.968 (0.520)	28.802 (1.304)	27.730 (1.406)	-0.830 (-1.315)	0.868 (0.801)	-0.873 (-1.209)	-0.870 (-1.378)
Cash	29.087*** (2.759)	12.379 (1.549)	38.590** (2.308)	28.955*** (2.751)	2.524*** (7.489)	2.512*** (5.957)	2.355*** (4.688)	2.507*** (7.462)
BDR	-7.130 (-1.367)	5.264 (0.771)	-7.306 (-1.179)	-7.051 (-1.357)	2.699*** (6.330)	3.949*** (6.467)	2.198*** (4.339)	2.696*** (6.354)
HARD	-1.861 (-0.831)	3.361 (1.483)	-3.315 (-0.897)	-1.796 (-0.805)	0.185*** (3.361)	0.257*** (2.774)	0.182*** (2.727)	0.189*** (3.444)
CSize	0.229 (0.469)	-0.265 (-0.519)	0.893 (1.216)	0.247 (0.506)	-0.008 (-0.532)	0.006 (0.295)	-0.013 (-0.561)	-0.007 (-0.445)
Disc.Exp	-22.166** (-1.963)	12.237 (1.086)	-41.125** (-2.388)	-24.716** (-2.133)	-0.697** (-1.988)	0.007 (0.014)	-1.129*** (-2.644)	-0.892** (-2.540)
Observations	7,290	3,213	4,075	7,290	7,290	3,213	4,075	7,290
Fixed Effects	Year×Ind	Year×Ind	Year×Ind	Year×Ind	Year×Ind	Year×Ind	Year×Ind	Year×Ind
R-squared	0.180	0.361	0.183	0.181	0.279	0.363	0.248	0.281

Note. Models 1–4 use Inventory Turnover as the dependent variable and include Profitability as a control. Models 5–8 use Profitability as the dependent variable and include Inventory Turnover as a control. “Pooled” denotes the full sample. “High Z” and “Low Z” are split by the year-specific cross-sectional median of lagged Altman Z-score. Models 4 and 8 use the full sample and additionally include lagged *NoDelay* (equals 1 if the firm has zero late payments beyond 30 days in the prior year). All models use OLS estimation with Year × Industry fixed effects. Robust standard errors are clustered by firm. t-statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ using two-tailed tests.

In the pooled sample, LPE is positively associated with subsequent inventory turnover (Model 1) and negatively associated with profitability (Model 5). These aggregate results are consistent with a potential flexibility-cost trade-off: firms that delay payments may gain short-term liquidity through payment timing, while incurring financial costs, such as forgone early-payment discounts and potential supplier price premiums (Chod 2017, Yang and Birge 2018).

However, Hypothesis 3 predicts that the inventory turnover association is mechanism-dependent. To test this prediction, we split the sample by lagged Altman Z-score. Table 9 (Models 2–3) reports the results. For financially healthy firms (Model 2), LPE has no significant association with inventory turnover ($\beta = -4.892$, $p > 0.10$), indicating that these firms maintain stable turnover despite delaying payments. This pattern is consistent with the flexibility mechanism, in which firms use

payment timing as a working-capital buffer without materially altering their inventory policy. On the other hand, for financially distressed firms (Model 3), LPE is positively and significantly associated with inventory turnover ($\beta = 23.237$, $p < 0.05$). Consistent with the distress mechanism described in Section 2.2, this pattern is consistent with accelerated inventory cycling under operational pressure (e.g., fire sales or inventory depletion) rather than operational efficiency.

We next examine whether late payments are associated with firm profitability. Models 6 and 7 of Table 9 show that LPE is negatively associated with the sales-to-COGS ratio for both financially healthy firms ($\beta = -0.533$, $p < 0.10$) and distressed firms ($\beta = -0.587$, $p < 0.05$), after controlling for inventory turnover. This result suggests that delayed payments are associated with margin erosion across both groups, consistent with the financial costs of forgone early-payment discounts or supplier price premiums. These results complement the financing-cost illustration in Section 4.1: while deferred payment generates gross financing-cost savings, the net association with profitability is negative, consistent with firms bearing real costs in exchange for short-term liquidity relief. For robustness, we provide a series of analyses in the Online Appendix.⁹

Collectively, these results reveal that the aggregate positive association between LPE and inventory turnover in the pooled sample conceals two distinct patterns. Financially healthy firms maintain stable turnover despite delaying payments, consistent with a flexibility-driven mechanism, whereas financially distressed firms exhibit elevated turnover associated with accelerated inventory cycling under operational pressure, consistent with a distress-driven mechanism. At the same time, the profitability results indicate that delayed payments are associated with financial costs across both groups. These findings contribute to the supply chain finance literature on trade credit (Haley and Higgins 1973, Kouvelis and Zhao 2012, Chod 2017, Yang and Birge 2018) by providing empirical evidence that the operational consequences of payment delays are heterogeneous and depend on firms' financial condition and the underlying mechanism governing payment behavior.

6.2. Flexibility Role of Late Payments

To further examine the flexibility role of payment timing, we analyze firms that do not rely on delayed payments. If short-term payment delays provide working-capital flexibility, firms that never delay payments should exhibit different operational patterns from those that use payment timing as a liquidity buffer. We therefore construct a *NoDelay* dummy variable, which equals 1 for firm-year observations with zero late payments exceeding 30 days and 0 otherwise. The 30-day threshold is intended to mitigate misclassification caused by routine operational payment schedules (e.g.,

⁹ Online Appendix OA4 decomposes LPE by duration and shows that the positive turnover association is driven by long-term delays, with short-term delays statistically insignificant once long-term delays are controlled for—consistent with the mechanism distinction developed above. Online Appendix OA5 addresses two alternative explanations, supplier rationing and non-operational deployment of liquidity.

fixed weekly or monthly payment cycles), which may generate minor delays without reflecting deliberate liquidity management. By excluding such cases, the dummy is intended to identify firms that effectively adhere to contractual payment terms.

We incorporate this variable into the performance regressions and report the results in Models 4 and 8 of Table 9. The coefficient on LPE remains statistically significant and of the same sign after including *NoDelay*, indicating that the baseline results are not solely attributable to firms that systematically avoid payment delays. The coefficient on *NoDelay* itself provides additional insight into the role of payment flexibility. In Model 4, firms that avoid payment delays exhibit lower inventory turnover ($\beta = -5.080$, $p < 0.1$), suggesting that the absence of payment timing as a liquidity buffer is associated with more conservative inventory cycling. This pattern is consistent with the flexibility mechanism described earlier, in which short-term payment delays allow firms to adjust working capital without materially altering inventory policies. In contrast, the profitability results (Model 8) indicate that avoiding payment delays does not eliminate the financial cost associated with trade credit timing. The coefficient on *NoDelay* is also negative and statistically significant ($\beta = -0.391$, $p < 0.01$), suggesting that strict adherence to payment terms does not necessarily restore profitability once operational conditions are accounted for.

7. Conclusion

Trade credit is one of the most important sources of short-term financing in supply chains, yet our understanding of how firms actually repay trade credit remains limited. Prior research has long recognized that information about trade credit provision is incomplete without observing customers' payment behavior (Ellingsen et al. 2016). However, due to the scarcity of detailed payment records, empirical studies have rarely been able to observe realized payment timing. Leveraging a unique dataset of payment records from Dun & Bradstreet, this study provides new empirical evidence on how firms adjust payment timing in practice. By observing realized payment behavior rather than relying on a binary view of repayment versus default, we document a more nuanced landscape in which payment delays vary continuously and serve as a discretionary instrument of liquidity management. Building on this empirical insight, we propose and test a unified framework that distinguishes between two mechanisms underlying payment delays: a flexibility mechanism, in which financially healthy firms adjust payment timing to manage working capital, and a distress mechanism, in which financially constrained firms delay payments due to operational pressure.

Our empirical findings reveal two key patterns. First, firms frequently delay trade credit payments as a means of transferring short-term liquidity pressure upstream to suppliers. However, these delays are typically bounded. Short-term delays are common in supply chains and rarely trigger enforcement actions, whereas extended delays are associated with a significantly higher

likelihood of litigation and the termination of buyer–supplier relationships. Second, the operational implications of delayed payments depend critically on firms’ financial conditions. Financially healthy firms maintain relatively stable inventory turnover despite delaying payments, consistent with the flexibility mechanism in which payment timing functions as a working-capital buffer. In contrast, financially distressed firms exhibit significantly higher inventory turnover, consistent with accelerated inventory cycling under operational pressure. At the same time, both groups exhibit lower profitability, suggesting that delayed payments are associated with financial costs such as forgone early-payment discounts or supplier price premiums. These findings also help reconcile our evidence with the credit contagion literature (Boissay and Gropp 2013), suggesting that late payments may signal distress primarily under financial constraints, while serving as liquidity flexibility for solvent firms.

These findings carry important managerial and policy implications. For managers, our results suggest that payment timing can serve as a useful instrument for short-term liquidity management, particularly when firms face temporary cash-flow fluctuations. At the same time, the evidence highlights the relational boundaries of this flexibility: persistent delays increase the risk of legal disputes and relationship termination, potentially undermining supplier stability and supply chain resilience. Managers should therefore view payment timing as a bounded flexibility tool rather than a costless source of financing. Moreover, our results help managers interpret payment delays from customers: delays occurring alongside stable inventory operations are more consistent with benign liquidity adjustments, while delays coupled with volatile inventory movements may signal distress-driven pressures and greater customer risk. For policymakers, the results suggest that regulatory attention should focus primarily on persistent and excessive payment delays that signal financial distress and threaten supply chain stability. Short-term payment adjustments, in contrast, often reflect routine liquidity management and may play a constructive role in coordinating cash flows across supply chain partners.

Finally, our findings open several avenues for future research. From an empirical perspective, the availability of detailed payment data provides opportunities to further examine liquidity transmission across supply chains, the propagation of financial distress through trade credit relationships, and the risk exposure of supplier firms, which recent evidence shows depends jointly on supplier leverage and horizontal market position (Ning et al. 2026). From a theoretical perspective, our results highlight an important gap in existing analytical models of trade credit. Much of the current literature treats trade credit repayment as a binary outcome—either full repayment according to contractual terms or outright default. Our evidence suggests that payment timing itself constitutes an important strategic instrument, allowing firms to adjust liquidity while preserving supply chain relationships within tolerated bounds. Incorporating payment timing and relational tolerance into

analytical models could yield new insights into optimal trade credit policies, inventory decisions, and supply chain coordination under financial constraints. More broadly, by documenting how payment timing varies within relational limits, this study positions trade credit not merely as a financing instrument, but as a mechanism of operational liquidity management embedded in supply chain relationships.

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Appendix A: Descriptive Summary of D&B Dataset

This appendix provides a detailed descriptive summary of the D&B dataset. Our D&B data sample spans from November 2004 to November 2016. However, as we use firm-year as our unit of analysis, we restrict the data to the full years from 2005 to 2015 to ensure consistency and comparability across observations. We next merge this dataset with firm financial information in Compustat. To illustrate the granular structure of the D&B dataset, we present several example records in Table A1, in which the variable definition is shown in Table A2. Each row is a single monthly observation for one firm (identified by CUSIP and four-digit SIC code). For instance, the first record shows a trucking firm (SIC 4212) in November 2004 with 210 total payment experiences, of which 83 are classified as late—broken down by duration from 30 days to over 180 days—and 2 as negative (trade credit that is highly unlikely to be recovered by the supplier, including trade credit that is in bad debt, under litigation, refusal to continue extending terms, under formal complaints, and perceived as not recoverable due to insufficient funds). The remaining records illustrate the variation across industries and time: an e-commerce retailer (SIC 5961) with 73 out of 175 experiences late, a food manufacturer (SIC 2026) with 146 out of 563 late, and a restaurant operator (SIC 5812) with only 3 out of 63 late. To construct our final sample, we aggregate the monthly records to the firm-year level and use CUSIP to match the D&B dataset with Compustat financials.

Table A1 Example Records from the D&B Dataset

Firm Identifiers				Payment Summary			Late Payment Breakdown by Duration						
<i>CUSIP</i>	<i>SIC</i>	<i>Year</i>	<i>Month</i>	<i>total_exp</i>	<i>nbr_disc</i>	<i>late_exp</i>	<i>30d</i>	<i>60d</i>	<i>90d</i>	<i>120d</i>	<i>180d</i>	<i>>180d</i>	<i>negative</i>
15349110	4212	2004	11	210	48	83	40	19	17	4	1	0	2
69037010	5961	2005	11	175	47	73	42	14	11	2	1	0	3
24237020	2026	2008	05	563	102	146	90	29	15	5	3	1	3
38214087	5812	2011	04	63	4	3	1	0	0	0	1	0	1

Note. Each row is a single monthly observation from the D&B trade payment dataset. *total_exp* is the total number of supplier payment experiences; *nbr_disc* is the number of suppliers with trade-credit discount terms; and *late_exp* is the sum of all late categories. The last seven columns break down late payments by duration: *30d* through *>180d* report the number of experiences late by each duration category, and *negative* reports experiences classified as negative by the reporting supplier.

Table A2 Variables in D&B Trade Credit Payment Data at the Firm-Year Level.

Variable	Definition
$total_exp_{i,t}$	Number of suppliers with trade credit relationships
$nbr_disc_{i,t}$	Number of suppliers with trade credit discount terms
$late_30d_{i,t}$	Number of suppliers with slow trade credit overdue for 1-30 days
$late_60d_{i,t}$	Number of suppliers with slow trade credit overdue for 31-60 days
$late_90d_{i,t}$	Number of suppliers with slow trade credit overdue for 61-90 days
$late_120d_{i,t}$	Number of suppliers with slow trade credit overdue for 91-120 days
$late_180d_{i,t}$	Number of suppliers with slow trade credit overdue for 121-180 days
$late_over180d_{i,t}$	Number of suppliers with slow trade credit overdue for over 181 days
$late_negative_{i,t}$	Number of suppliers with negative trade credit
$late_exp_{i,t}$	Number of suppliers with slow or negative trade credit

Notes: For consistency, we have $late_exp_{i,t} = late_30d_{i,t} + late_60d_{i,t} + late_90d_{i,t} + late_120d_{i,t} + late_180d_{i,t} + late_over180d_{i,t} + late_negative_{i,t}$